L2.5 Curve Sketching

Recall: L2.4 Limits at Infinity; Horizontal Asymptotes

Name ________________________________ **Solo – 9 minutes**

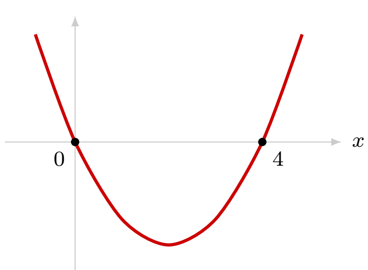
1. Evaluate $\lim_{x\to\infty} \dfrac{5x^2+3x-1}{2x^2-7}$. Show the division step.

2. Evaluate $\lim_{x\to-\infty} \dfrac{\sqrt{4x^2+1}}{3x-2}$. Careful — the sign is the whole problem.

3. Everything below is true of the same function.

D	all real numbers
I	zeroes at $x = 0$ and $x = 4$, and nowhere else; y-intercept 0
A	no vertical or horizontal asymptotes. **EB**: $f(x) \to +\infty$ at both ends
S	neither even nor odd

Both sketches below fit *every* row. Which is the real graph — and what would settle it?



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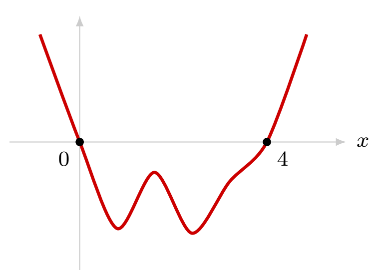

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